

Bobby Chilami

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Summary

Senior Software Engineer and Team Lead with 4+ years of experience building **scalable backend systems, microservices, REST APIs, cloud-native applications, and AI solutions**. Strong in **Java, Spring Boot, Python, TypeScript/Node.js, SQL, AWS, Docker, and CI/CD**, with hands-on experience in system design, distributed systems, performance optimization, API security, and enterprise integrations. Experienced in **LLM/RAG applications, AI agents, vector search, tool calling, and natural-language-to-SQL**, with proven ownership of end-to-end systems and engineering teams serving **millions of users**.

Technical Skills

- **Languages:** Java, Python, TypeScript, JavaScript, C#, SQL, Bash
- **Backend & APIs:** Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, FastAPI, Node.js, Express.js, REST APIs, Microservices, JWT, OAuth 2.0
- **AI/LLM:** Generative AI, LLMs, RAG, Vector Search, Embeddings, Prompt Engineering, AI Agents, Tool/Function Calling, Natural Language-to-SQL, LLM APIs, LangChain, LlamaIndex, Semantic Kernel, Qwen 2.5 Coder 7B
- **Databases:** MySQL, PostgreSQL, Microsoft SQL Server, MongoDB, Redis, pgvector, Pinecone, Weaviate, Milvus, Database Design, Query Optimization, Indexing
- **Cloud & DevOps:** AWS (EC2, S3, RDS, CloudWatch), Docker, Kubernetes, Jenkins, CI/CD, Load Balancing, Auto Scaling, Linux, Git, GitHub
- **Engineering:** Data Structures & Algorithms, System Design, Distributed Systems, Scalable Architecture, Event-Driven Architecture, Concurrency, Performance Optimization, Unit/Integration/API Testing, Observability, Monitoring, Error Handling

Experience

Flixir Solutions

Pune, India

Team Lead / Software Engineer

Sept 2022 – Present

- Led multiple engineering teams across **SSC/HSC, Skoda manufacturing, and AI initiatives**, coordinating developers and testers across architecture, requirements, development, code reviews, testing, CI/CD, deployment, production support, mentoring, technical interviews, and stakeholder coordination.
- Architected and led development of a Maharashtra State Board SSC/HSC platform serving approximately **30 lakh users**, owning system architecture, database design, REST APIs, authentication/security, scheduled workflows, AWS infrastructure, scalability, and production operations.
- Designed scalable **microservices and RESTful backend services** processing **1M+ records and supporting 5K–10K concurrent users**, applying distributed-systems principles, fault isolation, API design, and performance engineering.
- Designed API-driven **enterprise integrations** connecting business applications, reporting systems, administrative platforms, data sources, and automated workflows with robust validation, error handling, and reliable data exchange.
- Optimized **MySQL queries, indexing, JSON operations, database workflows, and Redis caching**, improving API response times by approximately **30–50%** and reducing database load for high-volume workloads.
- Deployed and operated production systems on **AWS EC2, RDS, S3, and CloudWatch**, implementing load balancing, auto-scaling, monitoring, and **Jenkins-based CI/CD** for automated build and deployment workflows.
- Led development of manufacturing software deployed across **2 production lines and 24 stations**, supporting 6 engine types and 50+ operators while integrating enterprise applications with Siemens S7 PLCs and shop-floor equipment; provided hands-on production troubleshooting and root-cause analysis.

Acty System India, Pvt. Ltd.

Software Engineer

India

May 2022 – Sept 2022

- Developed and maintained enterprise software using **C and .NET Framework**, contributing to backend development, application architecture, API integration, debugging, testing, and production delivery.
- Worked on database-driven enterprise workflows, applying **object-oriented programming, API integration, debugging, and software engineering best practices**.

QuestPond

Software Engineer Intern

Remote

Nov 2021 – Dec 2021

- Developed **Python machine-learning models** using Linear Regression and seasonal decomposition to forecast equipment parameters and support predictive maintenance analysis.
- Built **C and SQL Server-backed analytical workflows** and Angular visualizations for equipment health, predicted parameters, network diagrams, and maintenance insights.

Projects

Sentinel – AI-Powered Database Agent | *Python, FastAPI, RAG, LLM, Qwen 2.5 Coder 7B, Vector Search, React*

- Built a local-first **AI database assistant** enabling natural-language interaction with structured and semi-/non-structured data without requiring SQL or schema knowledge.
- Designed an end-to-end **RAG pipeline** for database metadata ingestion, schema processing, semantic search, context retrieval, and LLM-assisted SQL generation.
- Integrated **Qwen 2.5 Coder 7B** for local inference and natural-language-to-SQL generation, with on-device processing for privacy-sensitive environments.
- Developed modular **FastAPI services** and a React/TypeScript frontend, incorporating tool calling, structured outputs, query validation, guardrails, error handling, and extensible agentic workflows.

Voice – AI-Powered Automated Tutor | *Python, LLM, AI Agents, Web Search, NLP*

- Developed an **AI agent workflow** combining web search, information retrieval, contextual processing, and LLM-based response generation for an automated tutoring application.

TranscriberHub – Multilingual AI Transcription Platform | *Python, FastAPI, LLM/AI, REST APIs*

- Built a **FastAPI backend and REST APIs** for multilingual YouTube transcription, supporting both caption-based and audio-based transcription workflows.
- Designed an extensible **asynchronous processing architecture** for multilingual audio processing and automated transcript-to-audio generation.

Education

Sant Gajanan Maharaj College of Engineering

B.Tech in Computer Science and Engineering

Kolhapur, Maharashtra

2018 – 2022

- **Academic Performance:** 87% Final Year – First Class with Distinction; 77% Overall
- **Achievements:** Best Outgoing Student (2022), Circle Hackathon Final Phase (Top 10%), 2nd Place – Coding Competition
- **Leadership:** Conducted programming workshops for 100+ students and mentored hackathon teams